31 Advanced configuration of ASP.NET Core

This chapter covers

- Building custom middleware
- Using dependency injection (DI) services in IOptions configuration
- Replacing the built-in DI container with a third-party container

When you're building apps with ASP.NET Core, most of your creativity and specialization go into the services and models that make up your business logic and the Razor Pages and APIs that expose them. Eventually, however, you're likely to find that you can't quite achieve a desired feature using the components that come out of the box. At that point, you may need to look to more complex uses of the built- in features.

This chapter shows some of the ways you can customize cross-cutting parts of your application, such as your DI container or your middleware pipeline. These approaches are particularly useful if you're coming from a legacy application or are working on an existing project, and you want to continue to use the patterns and libraries you're familiar with.

We'll start by looking at the middleware pipeline. You saw how to build pipelines by piecing together existing middleware in chapter 4, but in this chapter you'll create your own custom middleware. You'll explore the basic middleware constructs of the `Map`, `Use`, and `Run` methods and learn how to create standalone middleware classes. You'll use these to build middleware components that can add headers to all your responses as well as